

AMAR DHILLON

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Education

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics; GPA: 3.5

May 2028
College Park, MD

Experience

Amazon Web Services

Sep 2026 – Dec 2026

Software Development Engineer Intern

Redmond, WA

- Researching and implementing Java-based utilization optimizations for Amazon Redshift (in progress)

Capital One

Jun 2026 – Aug 2026

Software Engineering Intern

Richmond, VA

- Rebuilt a legacy test-data workflow builder UI in React, TypeScript, and Electron with a React-Flow editor and Jest-tested validation; **90%** of surveyed users preferred it over the old tool
- Built a precomputed tree-style search index over popular workflows from a **270K+ row** PostgreSQL table (Python/SQLAlchemy, Redis cache) to minimize DB queries under access limits
- Cut LLM workflow generation time **65%** to under 15s by decomposing a Claude skill into sub-skills backed by Python search/validation tools, guided by CloudWatch + OpenSearch latency logs

UMD App Development Contracting (Client: Amazon)

Jan 2026 – Present

Software Engineer (Contract)

College Park, MD

- Shipped a space-research tool for Amazon Leo, now adopted into Leo's ecosystem; deployed on Docker + AWS Fargate via GitHub Actions, with AWS CDK (TypeScript) IaC for easy onboarding
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by moving it onto Kinesis streams feeding Lambda processors
- Automated fine-tuning of a Bedrock source-relevancy model with EventBridge-scheduled runs and decision gates that skip redundant retraining, plus X-Ray tracing and SNS failure alerts

UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

Machine Learning Research Intern

College Park, MD

- Developing automated training for PhysTwin, a physics-based digital twin, on a PhD-led team targeting ICLR
- Engineered a Python pipeline rendering AnyGrasp multi-camera grasp candidates as Open3D overlays for a Qwen3-VL-8B vision-language model, letting researchers isolate scene components
- Shortened research-environment setup time **30%** with uv dependency pinning and Docker; added psutil time/memory profiling and Pytest suites to find bottlenecks

TerpLabs

Jun 2025 – Present

Co-Founder, Software Engineer

College Park, MD

- Co-founded a **110+ member** campus product studio that has shipped 5 live products, run with Jira and GitHub
- Wrote Prisma SQL queries and Redis query caching for Tortuga to reduce database load and latency; the class scheduler drew **20,000+ visits** in its first 30 days
- Building a RAG meal-suggestion feature for the official UMD app (GPT-4o mini, ChromaDB + SentenceBERT search); **93%** of generated meals rated satisfactory by internal testers

Projects

IEX Matching Engine Simulation | C++, STL, POSIX

Jul 2026 – Aug 2026

- Implemented a C++ order-book simulator replaying historical IEX data, with a contract-based plug-in interface so alternative approaches swap in under enforced correctness checks
- Designed paginated hash-based order-book storage with an in-memory LFU cache for hot tickers during simulation, checked for drift against IEX historical book data
- Benchmarked a time-based adaptive LSM store against a B+ tree for simulation logs, measuring throughput and latency via rdtscp and hardware utilization via Linux perf counters

Technical Skills

Languages: Python, Java, C++, C, TypeScript/JavaScript, SQL

AWS: Lambda, Kinesis, Fargate/ECS, API Gateway, EventBridge, Bedrock, CDK, CloudWatch, X-Ray, SNS

Backend & Data: Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

AI/ML & Tools: PyTorch, RAG, Open3D, Docker, GitHub Actions, Linux, Pytest, Jest, uv, perf

Certification: AWS Certified Developer – Associate (Nov 2024 – Nov 2027)